

Morgan Huberty

[Email](#) | [LinkedIn](#) | [Website](#) - <https://mhubert3.github.io/>

Education

Johns Hopkins University

Expected May 2027

Bachelor of Science in Computer Engineering

Relevant Coursework: Data Structures, Computer System Fundamentals, Signals & Systems, FPGA Synthesis

Projects

Big Integers

C++

- Implemented a BigInt class (vector<uint64_t> magnitude, bool sign) supporting arbitrary-precision add, sub, mul, div, bit ops, and hex/dec conversion, with comprehensive unit tests

Key/Value Store

C++

- Built a Redis-like in-memory K/V store with custom message serialization, multithreaded server (auto-commit & transactions with table-level locking), client utilities, and unit tests

Cache Simulator

C, C++

- Simulated a configurable set-associative cache (size, block, LRU/FIFO, write policies) that outputs hit/miss and cycle counts from benchmark traces

Parallel Merge Sort

C

- Implemented fork/join parallel merge sort on memory-mapped 64-bit integer files, with threshold-based sequential fallback, robust child-process management, and performance tuning experiments

Experience

#iVoted Impact Concerts at SNF Agora Institute

May 2025 - Present

Data Science Researcher

- Analyzed music-listener and voter-demographic datasets (RedistrictingDataHub, Geocorr APIs) to inform #iVoted interventions
- Built projection dashboard for voter registrations using Python, Excel, and tracked data from #iVoted interventions

National Society of Black Engineers

September 2023 - Present

Vice President; previous Finance Chair

- Managed financial planning for chapter activities, ensuring allocating and stewarding a \$9K annual budget
- Developed marketing materials to promote member engagement and outreach by leveraging digital platforms
- Coordinated fundraising and sponsorship efforts for \$3500 worth of income

VEX Robotics

November 2019 - March 2023

Robotics Designer

- Designed and programmed competition-grade robotics platforms using C++ and Python
- Performed hardware bring-up (calibration; sensor integration) and basic circuit assembly (motor controllers; wiring harnesses)
- Collaborated with team members in maker-space environment, developing tools and strategies to optimize robot performance
- Troubleshoot mechanical and electrical failures, documented design iterations and updated CAD models for 3D prints

Skills

Programming Languages: C, C++, Python (NumPy, Matplotlib, Pandas), MATLAB, Bash

Familiarity with VHDL, R, x86-64 Assembly (pixel graphics project)

Computer Applications: Microsoft Suite, Google Suite, Google Sheets, Excel, Slack, Canva, Docker, Vercel, Git, Tinkercad, Fusion360

Familiarity with KiCad, AWS, Google Cloud, Linux (Ubuntu), PyTorch, TensorFlow